

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

ANS:-

Laptop Won't Power On

Step	Troubleshooting Action	What I am checking
1	Check the power adapter and cable. Try another power outlet.	Checks whether the charger, cable, or wall socket is faulty.
2	Check the charging indicator on the laptop.	Confirms whether the laptop is receiving power from the adapter.
3	Disconnect the charger and remove the battery if removable, then hold the power button for 15–20 seconds. Reconnect power and try again.	Clears residual electrical charge and checks for a temporary power-state problem.
4	Try turning on the laptop using only the AC adapter, if the battery is removable.	Determines whether a faulty battery is preventing startup.
5	Check for signs of hardware failure such as no LEDs, unusual sounds, or a burning smell.	Helps identify possible motherboard, power circuit, or other internal hardware problems.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue. Hint: Consider checking both hardware connections and basic display settings.

ANS:-

Troubleshooting flow:

1. Check the monitor power

- Make sure the monitor is switched on and the power LED is lit.
- Check the monitor's power cable and power outlet.

2. Check the display cable

- Make sure the HDMI/DisplayPort/VGA cable is firmly connected to both the computer and monitor.
- If possible, try another cable.

3. Check the monitor input/source

- Use the monitor's buttons to select the correct input, such as **HDMI, DisplayPort, or VGA**.

4. Check whether the computer is actually producing a display signal

- Listen for startup sounds and check keyboard indicators such as Caps Lock.
- If available, connect the computer to another monitor.

5. Check Windows display settings

- Press **Windows + P** and select **PC screen only** or **Duplicate**.
- This checks whether the display output was accidentally switched to another screen.

6. Check the graphics hardware

- If the computer has a dedicated graphics card, make sure it is properly seated and the monitor cable is connected to the correct graphics output.

7. Restart the system

- Restart the computer and monitor and test again.

Conclusion: If another monitor works, the original monitor or cable is likely the problem. If neither monitor displays anything, further investigation of the graphics card, RAM, motherboard, or system startup is required.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

ANS:-

Wireless Mouse Is Unresponsive but Keyboard Works

To isolate the problem:

1. Check the mouse itself

- Turn the mouse off and on.
- Replace or recharge its batteries.
- Check whether the optical sensor is clean.
- Test the mouse on another computer.

2. Check the USB receiver

- Remove and reconnect the wireless receiver.
- Try plugging it into a different USB port.
- If the mouse works on another USB port, the original USB port may be faulty.

3. Check USB ports

- Connect another USB device, such as a flash drive, to the suspected port.
- If the other device also fails, the USB port may be the problem.

4. Check system settings/drivers

- Open **Device Manager** → **Mice and other pointing devices**.
- Check for a warning symbol or driver problem.
- Update or reinstall the mouse driver if necessary.

5. Check wireless connection

- For Bluetooth mice, make sure Bluetooth is enabled and the mouse is properly paired.
- For USB-receiver mice, check that the receiver is recognized by the system.

6. Final isolation

- Mouse fails on multiple computers → **mouse is likely faulty**.
- Mouse works on another computer → **system/settings/driver issue**.
- Other USB devices fail on one port → **USB port issue**.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one. Hint: Think about fans, dust, and thermal paste.

ANS:-

Gaming PC Overheating and Shutting Down

Possible Hardware Cause	How to Check
1. Faulty or blocked cooling fan	Check whether the CPU/GPU fans are spinning during operation. A fan that is stopped, noisy, or spinning slowly may need cleaning or replacement.
2. Dust buildup	Inspect the CPU heatsink, GPU, case fans, and air vents. Excessive dust restricts airflow and prevents heat from escaping. Clean the components using appropriate compressed air while the PC is powered off.
3. Dried or improperly applied thermal paste	If the CPU temperature is unusually high despite working fans and clean airflow, inspect the CPU cooler. Old or dried thermal paste may need to be removed and replaced with fresh thermal paste.

Note: Overheating can trigger automatic thermal shutdowns to protect the CPU/GPU from damage.